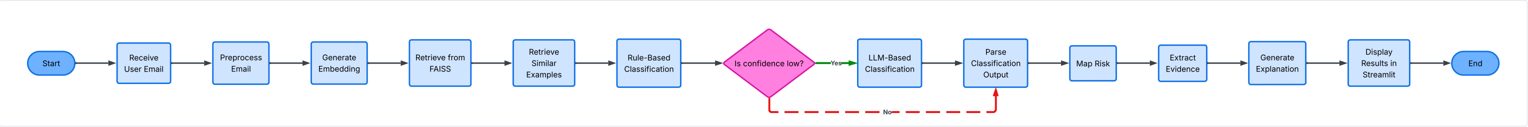
Email Risk Classification Process

Email Compliance Classification Process
Scope: Automated pipeline from receiving a user email through preprocessing, retrieval, classification, risk mapping, evidence and explanation generation, to displaying results in Streamlit.

Diagram key
- Terminator
- Process
- Decision



Email Risk Classification Pipeline

Diagram information
This diagram represents an automated email risk classification pipeline: starting from receiving a user email, preprocessing, embedding generation, retrieval of similar examples via FAISS, rule-based classification with a confidence check, fallback to LLM-based classification if confidence is low, mapping risk, extracting evidence, generating explanations, and displaying results in a Streamlit UI.

Email Risk Classification Pipeline SOP
1. Start the pipeline by receiving the user email.
2. Preprocess the email (clean text, normalize, remove signatures/headers).
3. Generate embedding for the preprocessed email.
4. Retrieve relevant vectors from FAISS using the generated embedding.
5. Retrieve similar examples and context from the FAISS results.
6. Apply rule-based classification on the email using retrieved context.
7. Evaluate classification confidence.
    7.1. If confidence is low, invoke LLM-based classification using the email and retrieved examples.
    7.2. If confidence is sufficient, proceed with the rule-based classification output.
8. Parse the final classification output into structured labels and metadata.
9. Map classification labels to risk levels and categories.
10. Extract supporting evidence from the email and retrieved examples that justify the classification.
11. Generate a human-readable explanation that summarizes the reasoning and evidence.
12. Display classification, risk mapping, evidence, and explanation in the Streamlit interface.
13. End the pipeline.